

1.

a)integer: its is data type which used integer number it in such as 2,3,4,1,0 there is no decimal number in it .

b)float :float uses decimal number in it but up to certain fix digit only unlike large decimal

c)string:string data type is represented by str ,its uses the str keyword ,it used for character alphumneral which is shown into quotations

d)boolean(bool):it only show true output either false for 0 or true for 1 and boolean datatype is used to check wheter result is false or true

2)

python program to create three variable

```
name ="Gargi"  
age=20  
city="Jaipur"  
print(f"Name:{name}\nAge:{age}\nCity:{city}\n")
```

program

```
Name:Gargi  
Age:20  
City:Jaipur
```

output

```
enter your name :tanvi  
TANVI  
5
```

```
name=input("enter your name :")  
print(name.upper())  
print(len(name))
```

3)

4)upper():it make all the letter in uppercas

split():it divide string into substring

replace():it is used gernlly in list and all to replace one element with another

strip():it removes whitespaccess

Capitalize :it captitalize the letter first generally

```
['mango', 'banana', 'kiwai', 'papaya', 'orange']  
mango  
orange  
5
```

5) `== Code Execution Successful ==`

```
1 list=["mango","banana","kiwai","papaya" ,"orange"]  
2 print(list)  
3 print(list[0])  
4 print(list[4])  
5 print(len(list))
```

```
[10, 20, 30, 40, 50]  
None  
None  
[10, 30, 40, 50, 60]
```

6) `== Code Execution Successful ==`

```
list=[10,20,30,40,50]  
print(list)  
print(list.append(60))  
print(list.remove(20))  
print(list)
```

7)artificial integgence is machine integgnce it can solve complex problems easily ,

Ai is very important in today s world for several tasks and information

Applications of ai are

Medical:in hospital ai system can be used whihc is very helpful

Ecommerce =in online commercial business ai helps a lot

Banking and finance :ai is also very helpful in this

Data analytics :ai can also help with anylisis of data in large companies or small

8) chat gpt is very commonly used ai for prompt generation by normal user for several information

Google maps = it is also ai it uses data to predict the routes and all

calculator = it is not ai it uses only mathematical operations

Netflix recommendation = it is ai since it uses user data to recommend content further with the help of algorithm

Voice assistants = yeah they recognise voice and answer according to the user prompt hence it is also ai